Firms in Competitive Markets

EC 111 Principles of Microeconomics

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Reading

- These slides are not a substitute for reading Mankiw, Chapter 15, "Firms in Competitive Markets".
- Required reading: Mankiw, *Principles of Microeconomics*, 11th edition, Chapter 15, "Firms in Competitive Markets".

What you must know from this chapter

1. Overview of market structures.
2. Conditions necessary to achieve perfect competition.
3. Profit maximization (MR = MC).
4. Graphical representation of profits/losses.
5. Shut-down point – short-run and long-run.
6. Short-run supply curve.
7. Entry/exit.
8. Firm and industry equilibrium and changes to equilibrium.

Part 1

Market Structures and Price Taking

Four market structures

When can a seller choose its own price?

Definition: Market structure

The number of firms, whether their products are identical or differentiated, and how easy it is to enter or leave the market.

Structure	Main feature	Firm's pricing position
Perfect competition	Many sellers of identical goods	Price taker
Monopolistic competition	Many sellers of differentiated goods	Some say over price
Oligopoly	A few important competing firms	Decisions depend on rivals
Monopoly	One seller; no close substitutes; barriers	Faces market demand

Perfect competition

Definition: Perfect competition and price taking

- Many buyers and sellers, each too small to move the market price.
- Firms sell **identical** products.
- A **price taker** accepts the market price.
- In the long run, firms can enter and leave freely: no significant barriers.

Free entry permits ordinary startup costs, such as buying equipment. It rules out significant barriers, such as exclusive legal rights.

Why a competitive firm is a price taker

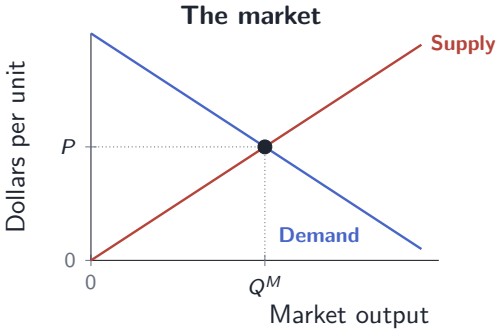
If products are identical and each seller is small, **then**:

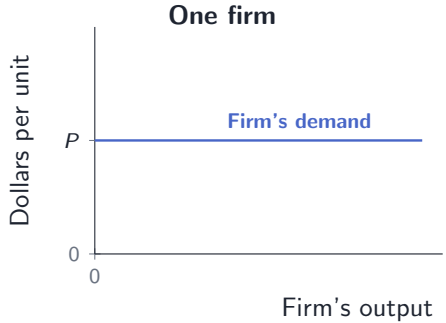
- Charging above the market price loses buyers to other sellers.
- Charging less cuts revenue per unit unnecessarily: the firm can sell its chosen output at the market price.
- If the market price is \$10, an identical pack offered at \$12 finds no buyers.

> **Finding**
> One firm's output cannot move the market price. Its demand curve is horizontal.

Market demand versus the firm's demand

P: market price; Q: one firm's output; Q^M: market output.





Finding

Market demand slopes down; the market sets P. One firm's demand is flat at P. If the market price changes, the firm's flat line moves with it.

Part 1 summary

- Many small sellers and identical goods make each firm a price taker.
- If market price changes, the firm faces a new horizontal demand line.
- Free entry means no significant entry barriers; ordinary startup costs remain.

Your turn: Check your understanding

The graph shows one small supplier.

1. Many sellers offer identical bolts. Best fit?

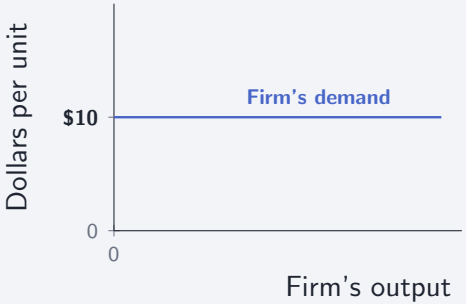
   A. Monopoly B. Perfect competition
   C. Oligopoly

2. Free entry means no equipment costs.

   A. True B. False

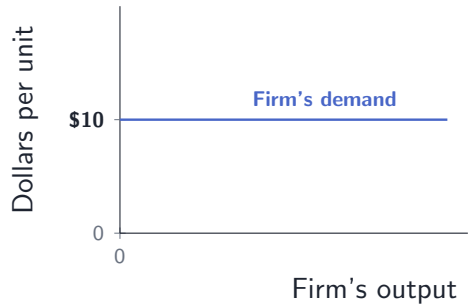
3. If market price rises, this demand line moves:

   A. Up B. Down C. Nowhere



Solution: Check your understanding

1. **B.** Many small sellers of identical goods fit perfect competition.
2. **B.** False: free entry concerns barriers, not ordinary equipment costs.
3. **A.** Up: the horizontal line moves up to the higher market price.



Check: Part 1

When firms are said to be price takers, it implies that if a firm raises its price,

A buyers will go elsewhere.

B buyers will pay the higher price in the short run.

C competitors will also raise their prices.

D firms in the industry will exercise market power.

Check: Part 1 – answer

When firms are said to be price takers, it implies that if a firm raises its price,

A **buyers will go elsewhere.**

B buyers will pay the higher price in the short run.

C competitors will also raise their prices.

D firms in the industry will exercise market power.

Finding

Many sellers offer an identical product at the market price. A firm that charges more than the market price loses its buyers to the other sellers. No single firm has market power.

Part 2

Price, Average Revenue and Marginal Revenue

Total, average and marginal revenue

Definition: Total revenue

$\text{TR} = P \times Q$: price times the quantity sold.

Definition: Average revenue

$\text{AR} = \text{TR}/Q$: revenue per unit sold, for $Q > 0$.

Definition: Marginal revenue

$\text{MR} = \Delta\text{TR}/\Delta Q$: the extra revenue from an extra unit sold. Δ means "change in".

A price taker: price is a parameter

A price taker cannot influence the market price. Whatever output it chooses, the price is fixed at $\bar{P}$: price is parametric.

- Total revenue: $\text{TR} = \bar{P} \times Q$.
- Average revenue: $\text{AR} = \dfrac{\text{TR}}{Q} = \dfrac{\bar{P} \times Q}{Q} = \bar{P}$.
- The firm can sell as many extra units as it likes at $\bar{P}$. Each extra unit adds $\bar{P}$ to revenue:

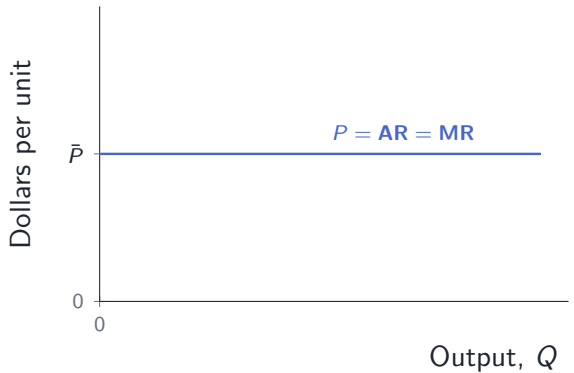
$$\text{MR} = \frac{\Delta \text{TR}}{\Delta Q} = \frac{\bar{P} \times \Delta Q}{\Delta Q} = \bar{P}.$$

Price equals average revenue equals marginal revenue

Finding

For a price-taking firm, $P = \text{AR} = \text{MR} = \bar{P}$.

The firm's demand curve is horizontal at $\bar{P}$. The same line is its AR curve and its MR curve.



Example: a price of \$10

Q	P	TR	AR	MR
1	\$10	$10 \times 1 = \$10$	$10/1 = \$10$	$(10-0)/1 = \$10$
2	\$10	$10 \times 2 = \$20$	$20/2 = \$10$	$(20-10)/1 = \$10$
3	\$10	$10 \times 3 = \$30$	$30/3 = \$10$	$(30-20)/1 = \$10$
4	\$10	$10 \times 4 = \$40$	$40/4 = \$10$	$(40-30)/1 = \$10$

Output rises one unit at a time, so $\Delta Q = 1$. Price, AR and MR are \$10 in every row.

Part 2 summary

- Total revenue is price times output: $\text{TR} = P \times Q$.
- Average revenue is revenue per unit: $\text{AR} = \text{TR}/Q$.
- Marginal revenue is the extra revenue per extra unit: $\text{MR} = \Delta\text{TR}/\Delta Q$.
- For a perfectly competitive firm, $P = \bar{P}$: price is parametric. The firm cannot influence price.
- So $P = \text{AR} = \text{MR}$.

Your turn: Check your understanding

Each unit sells for \$6.

1. MR from the third unit is:
   A. \$3 B. \$6 C. \$18
2. At three units, both TR and MR are \$18.
   A. True B. False
3. If the price rises to \$8, AR becomes:
   A. \$2 B. \$6 C. \$8

Q	P	TR	AR	MR
0	\$6	\$0	—	—
1	\$6	\$6	\$6	\$6
2	\$6	\$12	\$6	\$6
3	\$6	\$18	\$6	\$6

$\Delta Q = 1$.

Solution: Check your understanding

1. **B.** MR $= (18 - 12)/1 = \$6$.
2. **B.** False: TR $= 6 \times 3 = \$18$, but MR $= \$6$.
3. **C.** At the new price, each unit earns \$8, so AR is \$8.

Q	P	TR	AR	MR
0	\$6	\$0	—	—
1	\$6	\$6	\$6	\$6
2	\$6	\$12	\$6	\$6
3	\$6	\$18	\$6	\$6

$\Delta Q = 1$.

Check: Part 2

Suppose a firm in a competitive market earned $1,000 in total revenue and had a marginal revenue of $10 for the last unit produced and sold. What is the average revenue per unit, and how many units were sold?

A $5 and 50 units

B $5 and 100 units

C $10 and 50 units

D $10 and 100 units

Check: Part 2 – answer

Total revenue \$1,000; marginal revenue of the last unit \$10. Average revenue, and units sold?

A \$5 and 50 units

B \$5 and 100 units

C \$10 and 50 units

D \$10 and 100 units

Finding

For a competitive firm $P = \text{AR} = \text{MR}$, so $P = \text{AR} = \$10$. Units sold: $Q = \text{TR}/P = 1{,}000/10 = 100$.

Part 3

Profit Maximization

Economic profit

A competitive firm sells as many identical units as it likes at the market price $\bar{P}$.

Definition: Economic profit

$\pi = \text{TR} - \text{TC}$. The symbol π means profit.
Profit is a function of output sold: $\pi(Q) = \text{TR}(Q) - \text{TC}(Q)$.
Total cost includes opportunity costs: the value of the owner's time, capital and other resources.

When is it worth producing more?

Raise output by ΔQ. Profit changes by

$$\Delta\pi = \Delta\text{TR} - \Delta\text{TC}.$$

- If extra revenue exceeds extra cost, profit rises: add the unit.
- If extra cost exceeds extra revenue, profit falls: cut output.

Finding

Increase output while extra revenue exceeds extra cost. Cut output when extra cost exceeds extra revenue.

Profit under competition

- Profit is total revenue minus total cost: $\pi = \text{TR} - \text{TC}$.
- Both depend on output: $\pi(Q) = \text{TR}(Q) - \text{TC}(Q)$.
- Under competition price is parametric (Part 2): $P(Q) = \bar{P}$ at every output.
- So $\text{TR}(Q) = \bar{P} \times Q$, and

$$\pi(Q) = \bar{P} \times Q - \text{TC}(Q).$$

Profit is highest where MR = MC

Definition: Marginal cost
$\text{MC} = \Delta\text{TC}(Q)/\Delta Q$: the extra cost of an extra unit.

- One more unit changes profit by $\Delta\pi = \text{MR} - \text{MC}$.
- If $\text{MR} > \text{MC}$, profit rises: increase output. If $\text{MC} > \text{MR}$, profit falls: cut output.
- Profit is maximized where $\text{MR} = \text{MC}$. There $\text{MR} - \text{MC} = 0$, so $\Delta\pi(Q) = 0$.

The rule, derived

Start from $\pi(Q) = \bar{P} \times Q - \text{TC}(Q)$ and set the change in profit to zero.

- Change output by ΔQ:

$$\frac{\Delta \pi(Q)}{\Delta Q} = \frac{\bar{P} \times \Delta Q}{\Delta Q} - \frac{\Delta \text{TC}(Q)}{\Delta Q} = 0.$$

- An extra unit means $\Delta Q = 1$, so

$$\bar{P} = \frac{\Delta \text{TC}(Q)}{\Delta Q} = \text{MC}.$$

Finding

The profit-maximizing rule for a competitive firm: set price equal to marginal cost, $P = \text{MC}$.

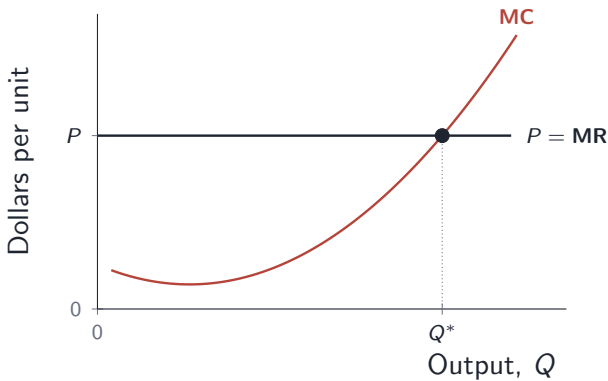
The rule on a graph

The firm faces $P = \text{MR}$.

Left of Q^*: $P > \text{MC}$. Each extra unit adds profit.

Right of Q^*: $P < \text{MC}$. Each extra unit loses profit.

Profit is greatest at Q^*, where $P = \text{MC}$.



Example: a firm's costs

$\text{TC}(Q) = F + \text{VC}(Q)$

$F = \text{FC} = \$16$: fixed cost.

$\text{VC}(Q) = Q^2 + 2Q$: variable cost.

Output rises one unit at a time: $\Delta Q = 1$.

MC in row Q is the cost of the move from $Q - 1$ to Q.

Each unit costs \$2 more than the one before: MC rises.

Q	FC	VC	TC	MC
0	\$16	\$0	\$16	—
1	\$16	\$3	\$19	\$3
2	\$16	\$8	\$24	\$5
3	\$16	\$15	\$31	\$7
4	\$16	\$24	\$40	\$9
5	\$16	\$35	\$51	\$11
6	\$16	\$48	\$64	\$13
7	\$16	\$63	\$79	\$15
8	\$16	\$80	\$96	\$17

Example: the best output at a price of \$13

Q	TC	MR	MC	TR	Profit
4	\$40	\$13	\$9	\$52	\$12
5	\$51	\$13	\$11	\$65	\$14
6	\$64	\$13	\$13	\$78	\$14
7	\$79	\$13	\$15	\$91	\$12
8	\$96	\$13	\$17	\$104	\$8

$\text{MR} = \bar{P} = \$13$; $\Delta Q = 1$; $\text{TR} = 13Q$.

Units 4 and 5: $\text{MC} < \text{MR}$. Profit rises.

Unit 6: $\text{MC} = \text{MR} = \$13$. $\Delta\pi = 0$.

Unit 7: $\text{MC} > \text{MR}$. Profit falls.

Best output 6, where $\text{MR} = \text{MC}$. Profit $78 - 64 = \$14$.

Why fixed cost drops out

Total cost: $\mathrm{TC}(Q) = F + \mathrm{VC}(Q)$. In the short run, F is owed either way.

- Profit: $\pi(Q) = \mathrm{TR}(Q) - F - \mathrm{VC}(Q)$.
- One more unit, $\Delta Q = 1$: $\Delta\pi = \Delta\mathrm{TR} - \Delta\mathrm{VC} - \Delta F$.
- F does not change with output, so $\Delta F = 0$:

$$\Delta\pi = \Delta\mathrm{TR} - \Delta\mathrm{VC} = \mathrm{MR} - \mathrm{MC}.$$

Finding

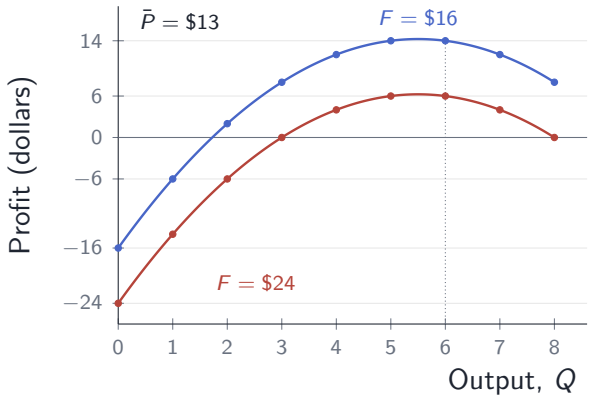
If variable costs and price stay unchanged, an unavoidable fixed cost cannot change the best short-run output. It changes profit.

A higher fixed cost lowers profit, not output (price \$13)

At $\bar{P} = \$13$, profit is $13Q - F - Q^2 - 2Q$. Initially $F = \$16$.

Fixed cost rises by \$8, to \$24. Every point falls by \$8.

Both curves are highest at the same output. At 6 units, profit falls from \$14 to \$6.



Part 3 summary

- Economic profit depends on output: $\pi(Q) = \text{TR}(Q) - \text{TC}(Q)$.
- Under perfect competition price is parametric, $P = \bar{P}$, so $\pi(Q) = \bar{P} \times Q - \text{TC}(Q)$.
- If $\text{MR} > \text{MC}$, increase output. If $\text{MC} > \text{MR}$, cut output.
- Profit is maximized where $\text{MR} = \text{MC}$: $\Delta\pi = \text{MR} - \text{MC} = 0$.
- For a price taker $\text{MR} = \bar{P}$, so the rule is $P = \text{MC}$.
- A change in fixed cost changes profit, not the profit-maximizing output.

Your turn: Check your understanding

The same firm, with the same costs.

1. At \$15, best output is:
   A. 6 B. 7 C. 8
2. At \$15, maximum profit is:
   A. \$26 B. \$79 C. \$105
3. At \$11, best output is:
   A. 4 B. 5 C. 6

Q	TC	MC	MR (\$15)	MR (\$11)
4	\$40	\$9	\$15	\$11
5	\$51	\$11	\$15	\$11
6	\$64	\$13	\$15	\$11
7	\$79	\$15	\$15	\$11
8	\$96	\$17	\$15	\$11

MR columns use prices \$15 and \$11.
$\Delta Q = 1$.

Solution: Check your understanding

1. **B.** Unit 7: MC $= \$15 =$ MR.
2. **A.** Profit $= 15 \times 7 - 79 = 105 - 79 = \$26$.
3. **B.** Unit 5: MC $= \$11 =$ MR.

Q	TC	MC	MR ($15)	MR ($11)
4	$40	$9	$15	$11
5	$51	$11	$15	$11
6	$64	$13	$15	$11
7	$79	$15	$15	$11
8	$96	$17	$15	$11

MR columns use prices $15 and $11.
$\Delta Q = 1$.

Check: Part 3

Farmer McDonald sells wheat to a broker in Kansas City, Missouri. Because the market for wheat is generally considered to be competitive, Farmer McDonald maximizes their profit by choosing

A to produce the quantity at which average variable cost is minimized.

B to produce the quantity at which average fixed cost is minimized.

C the quantity at which market price is equal to Farmer McDonald's marginal cost of production.

D the quantity at which market price exceeds Farmer McDonald's marginal cost of production by the greatest amount.

Check: Part 3 – answer

A competitive wheat farmer maximizes profit by choosing

A to produce the quantity at which average variable cost is minimized.

B to produce the quantity at which average fixed cost is minimized.

C the quantity at which market price is equal to Farmer McDonald's marginal cost of production.

D the quantity at which market price exceeds Farmer McDonald's marginal cost of production by the greatest amount.

Finding

The wheat market is competitive, so MR $= P$. Profit is maximized where MR $=$ MC, so where $P =$ MC. The largest gap between P and MC is on an early unit, not the profit-maximizing output.

Part 4

Profit and Loss on the Graph

Two ways to write profit

Definition: Average total cost

ATC $= \text{TC}/Q$: total economic cost per unit, for $Q > 0$.

Start from the definition of profit and use $P = \bar{P}$:

$$\begin{aligned}\pi &= \text{TR} - \text{TC} \\ &= \bar{P} \times Q - \text{TC} \\ &= \left(\bar{P} - \frac{\text{TC}}{Q}\right) \times Q \\ &= (\bar{P} - \text{ATC}) \times Q.\end{aligned}$$

Finding

Two ways to write the same profit: $\pi = \text{TR} - \text{TC}$, or $\pi = (\bar{P} - \text{ATC}) \times Q$. The second shows profit on a graph.

Profit as a rectangle

$\pi = (\bar{P} - \text{ATC}) \times Q$: profit per unit times the number of units.

- On a graph, profit is a rectangle: height $\bar{P} - \text{ATC}$, width Q.
- If $\bar{P} > \text{ATC}$, the firm makes a profit.
- If $\bar{P} < \text{ATC}$, the firm makes a loss. The rectangle's height is $\text{ATC} - \bar{P}$.
- If $\bar{P} = \text{ATC}$, economic profit is zero. There is no rectangle.

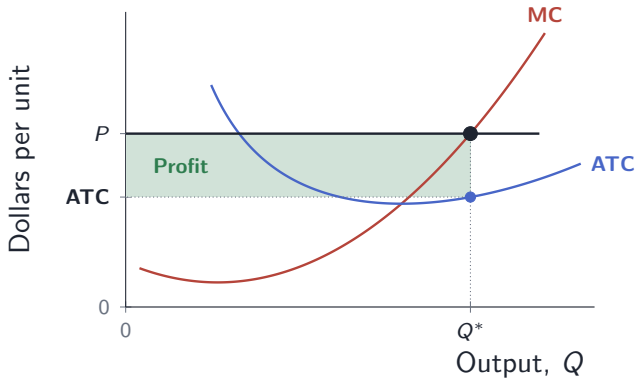
Profit on the graph

A general firm.

If $P >$ ATC at Q^*, the firm earns a profit.

Choose Q^* where $P =$ MC, then read ATC at that output.

Profit rectangle:
height $P -$ ATC, width Q^*.
Area $= (P -$ ATC$) \times Q^*$.



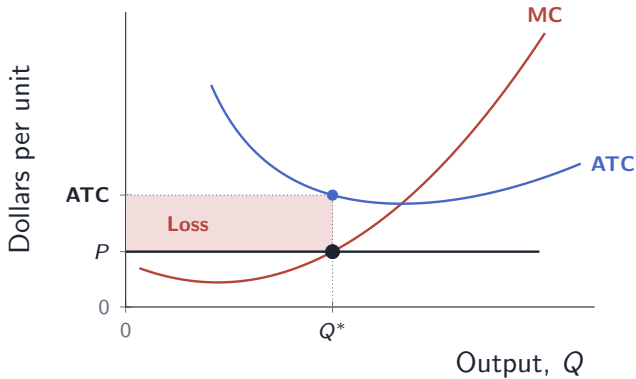
Loss on the graph

The same general firm.

If $P <$ ATC at Q^*, producing this output makes a loss.

Choose Q^* where $P =$ MC, then read ATC at that output.

Loss rectangle:
height ATC $-$ P, width Q^*.
Area $=$ (ATC $-$ P) $\times$ Q^*.



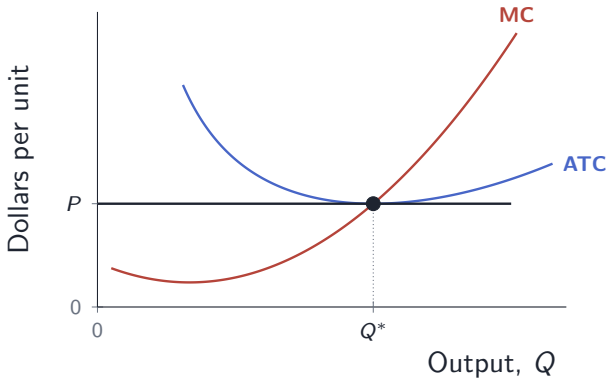
Zero profit on the graph

The same general firm.

If $P = \text{ATC}$ at Q^*, economic profit is zero.

$P = \text{MC} = \text{ATC}$ at minimum ATC: the break-even price.

The rectangle has zero height. Revenue exactly covers total economic cost.



Example: the running firm at two prices

Same firm as Part 3. Choose output where MR = MC; then compare $\bar{P}$ with ATC at that output.

$\bar{P} = \text{MR}$	MC	Q	TC	ATC	$(\bar{P} - \text{ATC}) \times Q$	TR − TC
\$17	\$17	8	\$96	$96/8 = \$12$	$(17 - 12) \times 8 = \$40$	$136 - 96 = \$40$
\$9	\$9	4	\$40	$40/4 = \$10$	$(9 - 10) \times 4 = -\$4$	$36 - 40 = -\$4$

Finding

Both ways give the same profit. At \$17, $\bar{P} >$ ATC: a profit of \$40. At \$9, $\bar{P} <$ ATC: a loss of \$4.

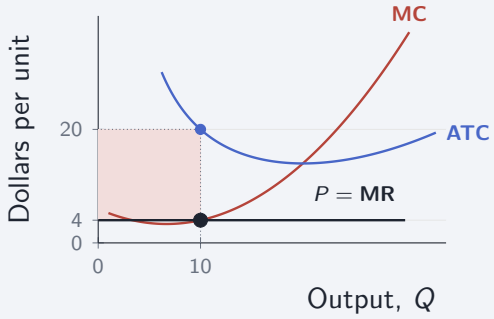
Part 4 summary

- Profit can be written two ways: $\pi = \text{TR} - \text{TC} = (\bar{P} - \text{ATC}) \times Q$.
- First choose output where MR = MC. Then compare $\bar{P}$ with ATC at that output.
- On a graph, profit is a rectangle: height $\bar{P} - \text{ATC}$, width Q. A loss has height $\text{ATC} - \bar{P}$.
- At the break-even price, $\bar{P}$ equals minimum ATC and economic profit is zero.

Your turn: Check your understanding

A general firm. Read the graph.

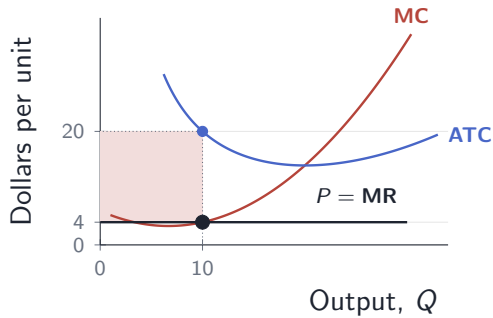
1. What is the best positive output?
   A. 4 B. 10 C. 20
2. What is economic profit?
   A. −\$160 B. \$40 C. \$200
3. The shaded rectangle is revenue.
   A. True B. False



Solution: Check your understanding

A general firm.

1. **B.** $Q = 10$, where $P = \text{MC}$ on rising MC.
2. **A.** $\pi = (4 - 20) \times 10 = -\$160$.
3. **B.** False: the rectangle shows the loss; revenue is $4 \times 10 = \$40$.



Check: Part 4

Assume a firm in a competitive industry is producing 800 units of output, and it sells each unit for \$6. Its average total cost is \$4. Its profit is

A $-\$1{,}600$.

B \$1,600.

C \$3,200.

D \$8,000.

Check: Part 4 – answer

800 units sold at \$6 each; average total cost \$4. Profit is

A $-\$1{,}600$.

B \$1,600.

C \$3,200.

D \$8,000.

Finding

$\pi = (P - \text{ATC}) \times Q = (6-4) \times 800 = \$1{,}600$. Check: $\text{TR} - \text{TC} = 4{,}800 - 3{,}200 = \$1{,}600$. \$3,200 is total cost, not profit.

Part 5

The Short-Run Supply Decision

Short-run costs

Definition: Short run
A period in which at least one input is fixed, so some costs cannot be avoided.

- Total cost: $\text{TC} = \text{FC} + \text{VC}$.
- Fixed cost FC does not change with output. It must be paid even at $Q = 0$.
- Variable cost VC rises with output Q.
- Average total cost $\text{ATC} = \text{TC}/Q$. Average variable cost $\text{AVC} = \text{VC}/Q$.

Profit and output: what we know

- Profit: $\pi = \bar{P} \times Q - \text{TC} = \bar{P} \times Q - \text{VC} - \text{FC}$. It can also be written $\pi = (\bar{P} - \text{ATC}) \times Q$.
- Output rule (Part 3): the firm takes $\bar{P}$ as given and produces where $\bar{P} = \text{MC}$.
- Profit or loss (Part 4): profit if $\bar{P} > \text{ATC}$, loss if $\bar{P} < \text{ATC}$, zero profit if $\bar{P} = \text{ATC}$.
- The break-even price is minimum ATC.

Finding

The question for this Part: if the firm makes a loss, should it produce at all?

Shutdown and sunk cost

Definition: Shutdown

A short-run decision to produce nothing for now. Fixed costs must still be paid.

Definition: Sunk cost

A cost already committed that cannot be recovered. In the short run, fixed cost is sunk. It should not affect what the firm does next.

The shutdown rule, derived

- Shut down: revenue and VC are zero, but FC is still paid. So $\pi = -\text{FC}$.
- Produce: $\pi = \bar{P} \times Q - \text{VC} - \text{FC}$.
- Producing is at least as good as shutting down when

$$\bar{P} \times Q - \text{VC} - \text{FC} \;\geq\; -\text{FC}.$$

- FC cancels: $\bar{P} \times Q \geq \text{VC}$. Divide by $Q > 0$: $\bar{P} \geq \text{VC}/Q = \text{AVC}$.

Finding

Shutdown rule: produce only if $\bar{P} \geq$ AVC. Shut down if $\bar{P} <$ minimum AVC. The shutdown price is minimum AVC.

Two ways to write the shutdown rule

Start from the shutdown condition and use $\text{TR} = \bar{P} \times Q$ and $\text{VC} = \text{AVC} \times Q$:

$$\begin{aligned} \text{TR} &< \text{VC} \\ \bar{P} \times Q &< \text{AVC} \times Q \\ \bar{P} &< \text{AVC} \qquad \text{(divide by } Q > 0\text{)} \\ (\bar{P} - \text{AVC}) \times Q &< 0. \end{aligned}$$

Finding

Shut down if $\text{TR} < \text{VC}$, which is the same as $\bar{P} < \text{AVC}$.

Example: operate or shut down?

The running firm: FC $= \$16$, VC $= Q^2 + 2Q$, so AVC $= Q + 2$. Minimum AVC is \$3, at $Q = 1$. Output is where $\bar{P} =$ MC.

$\bar{P}$	Q	TR	VC	Profit: produce	Profit: shut	Decision
\$9	4	\$36	\$24	$36 - 24 - 16 = -\$4$	$-\$16$	Produce: TR $>$ VC
\$3	1	\$3	\$3	$3 - 3 - 16 = -\$16$	$-\$16$	Shutdown price: $\bar{P} =$ min AVC
\$2	0	\$0	\$0	—	$-\$16$	Shut down: $\bar{P} <$ min AVC

Producing at a loss

Take a price with AVC $< \bar{P} <$ ATC.

- Because $\bar{P} >$ AVC, the firm produces.
- Because $\bar{P} <$ ATC, it makes a loss.
- Revenue covers all of VC and part of FC. Shutting down loses all of FC, so producing loses less.
- The running firm at \$9: TR \$36 covers VC \$24 and \$12 of the \$16 FC. It loses \$4 by producing and \$16 by shutting down.

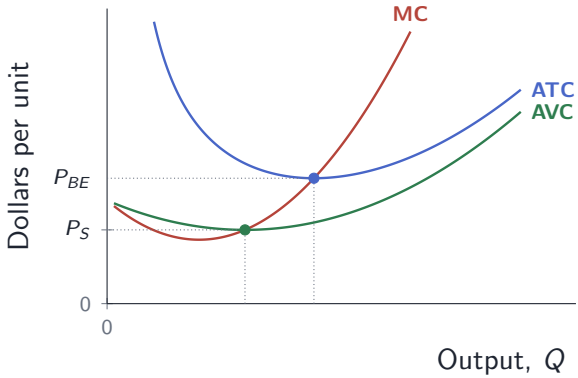
Finding

In the short run a firm with AVC $< \bar{P} <$ ATC stays in production, at a loss.

Shutdown and break-even prices on the graph

General U-shaped curves. MC meets minimum AVC at the shutdown price P_S.

MC crosses ATC at its lowest point: the break-even price P_{BE}.



Finding

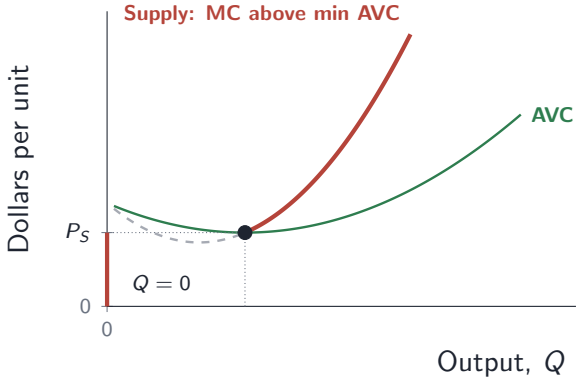
$\bar{P} < P_S$: shut down. $P_S < \bar{P} < P_{BE}$: produce at a loss. $\bar{P} > P_{BE}$: produce at a profit.

The firm's supply curve, derived

Output rule: at each price $\bar{P}$, produce where $\bar{P} = \text{MC}$.

Shutdown rule: produce only if $\bar{P} \geq$ minimum AVC (P_S).

Below P_S the firm shuts down and supplies zero.



Finding

The firm's short-run supply curve is its MC curve above minimum AVC: the upward-sloping part of MC.

Example: the supply schedule

The running firm. At each price: output where $\bar{P} = \text{MC}$; produce only if $\bar{P} \geq$ minimum AVC $= \$3$.

$\bar{P}$	Unit with MC $= \bar{P}$	AVC at that Q	Q supplied	Decision
\$2	—	—	0	Shut down: \$2 $<$ \$3
\$3	1	\$3	1	Shutdown price
\$9	4	\$6	4	Produce
\$13	6	\$8	6	Produce
\$17	8	\$10	8	Produce

Finding

A higher price moves the firm up its MC curve: it supplies more.

From one firm to market supply

Definition: Market supply

The horizontal sum of the firms' supply curves: at each price, add the quantities all firms supply.

100 identical firms, each with the schedule above.

$\bar{P}$	One firm	Market $= 100\times$ one firm
\$9	4	$100 \times 4 = 400$
\$13	6	$100 \times 6 = 600$
\$17	8	$100 \times 8 = 800$

In the short run the number of firms is fixed, so market supply slopes up because each firm's MC slopes up.

Part 5 summary

- Short run: $\text{TC} = \text{FC} + \text{VC}$. FC is paid even at $Q = 0$.
- Output rule: produce where $\bar{P} = \text{MC}$.
- Shutdown rule: produce only if $\bar{P} \geq \text{AVC}$. The shutdown price is minimum AVC.
- The firm's short-run supply curve is its MC curve above minimum AVC. Below minimum AVC it supplies zero.
- If $\text{AVC} < \bar{P} < \text{ATC}$, the firm produces at a loss: it loses less than FC.
- Market supply adds the firms' quantities at each price.

Your turn: Check your understanding

The running firm. Fixed cost is \$16, paid either way.

1. At \$9, the firm should:
   A. Produce B. Shut down
2. The shutdown price is:
   A. \$2 B. \$3 C. \$9
3. The firm's short-run supply curve is:
   A. Its AVC curve B. Its MC curve above minimum AVC C. Its ATC curve

$\bar{P}$	Q	TR	VC	AVC
\$9	4	\$36	\$24	\$6
\$3	1	\$3	\$3	\$3

Q: output where $\bar{P} = \text{MC}$.

Solution: Check your understanding

1. **A.** TR $= \$36 >$ VC $= \$24$. Profit from producing: $36 - 24 - 16 = -\$4$; from shutting down: $-\$16$.
2. **B.** Minimum AVC $= 3/1 = \$3$.
3. **B.** Output rule plus shutdown rule: MC above minimum AVC.

$\bar{P}$	Q	TR	VC	AVC
\$9	4	\$36	\$24	\$6
\$3	1	\$3	\$3	\$3

Q: output where $\bar{P} =$ MC.

Check: Part 5

For a competitive firm, which of the following represents the short-run condition for the firm to shut down?

A Shut down if TR $<$ TC

B Shut down if TR $<$ FC

C Shut down if $P <$ ATC

D Shut down if TR $<$ VC

Check: Part 5 – answer

For a competitive firm, which of the following represents the short-run condition for the firm to shut down?

A Shut down if TR $<$ TC

B Shut down if TR $<$ FC

C Shut down if $P <$ ATC

D Shut down if TR $<$ VC

Finding

FC is paid whether the firm produces or not, so it cancels. Produce only if TR $\geq$ VC; shut down if TR $<$ VC, the same as $(\bar{P} - \text{AVC}) \times Q < 0$. A and C describe a loss, which a firm can accept in the short run.

Part 6

Entry and Exit

The long run: exit

Definition: Long run

A period long enough to change all inputs. Nothing is fixed: FC $= 0$ and TC $=$ VC. Every cost can be avoided by leaving the industry (exit).

- In the short run the firm produced at a loss because FC had to be paid anyway.
- In the long run, leaving costs nothing. A firm with $\bar{P} <$ ATC loses money by staying and nothing by leaving, so it exits.

Finding

Long-run rule: stay only if $\bar{P} \geq$ minimum ATC. Exit if $\bar{P} <$ minimum ATC.

Entry and exit change market supply

If firms make a profit or loss, what changes? Assume identical firms, free entry and exit, and unchanged input prices.

If economic profit is positive

- New firms enter.
- Market supply shifts right.
- The price falls and profit shrinks.

If economic losses persist

- Some firms exit.
- Market supply shifts left.
- The price rises and losses shrink.

Finding

Entry and exit stop at zero economic profit: $P =$ minimum ATC.

Zero economic profit

Definition: Zero economic profit

Total revenue covers all economic costs, including the opportunity costs of the owner's time, capital and other resources.

- The salary given up to run the business is part of economic cost.
- Accounting profit can be positive when economic profit is zero.
- The owner earns the normal return available elsewhere. There is no extra economic profit to attract entry.

Part 6 summary

- If economic profit is positive, firms enter; supply shifts right and price falls.
- If losses persist, firms exit; supply shifts left and price rises.
- With identical costs and unchanged input prices, adjustment stops at zero economic profit. The owner still earns a normal return.

Your turn: Check your understanding

Free entry; input prices unchanged.

1. In case 1, entry moves market supply:
   A. Right B. Left C. Not at all
2. Zero economic profit means no reward for the owner's work.
   A. True B. False
3. In case 2, economic profit is:
   A. \$15 B. \$0 C. \$60

Case 1: identical firms	
Market price	\$12
Minimum ATC	\$10
Case 2: one owner	
Revenue	\$60
Payments to others	\$45
Forgone wage	\$15

Solution: Check your understanding

1. **A.** \$12 exceeds minimum ATC of \$10, so profit attracts entry.
2. **B.** False: economic cost includes the owner's normal return.
3. **B.** Economic profit $= 60 - 45 - 15 = \$0$; accounting profit is \$15.

Case 1: identical firms	
Market price	\$12
Minimum ATC	\$10
Case 2: one owner	
Revenue	\$60
Payments to others	\$45
Forgone wage	\$15

Check: Part 6

Profit-maximizing firms enter a competitive market when existing firms in that market have

A total revenues that exceed fixed costs.

B total revenues that exceed total variable costs.

C average total costs that exceed average revenue.

D average total costs that are less than market price.

Check: Part 6 – answer

Profit-maximizing firms enter a competitive market when existing firms in that market have

A total revenues that exceed fixed costs.

B total revenues that exceed total variable costs.

C average total costs that exceed average revenue.

D average total costs that are less than market price.

Finding

If $P >$ ATC, existing firms earn positive economic profit, $\pi = (P - \text{ATC}) \times Q > 0$. That profit attracts entry. B is the short-run condition for producing, not for entry.

Part 7

Firm and Industry Equilibrium, and Changes to Equilibrium

Short-run and long-run equilibrium

If demand changes, what happens now and after firms can enter or exit?

Definition: Short-run equilibrium

Market quantity demanded equals quantity supplied. Each firm maximizes profit at the market price. Firms can make profits or losses.

Definition: Long-run equilibrium

With identical firms, free entry and exit, and unchanged input prices: firms earn zero economic profit and produce at efficient scale.

Definition: Efficient scale

The output at which average total cost is lowest.

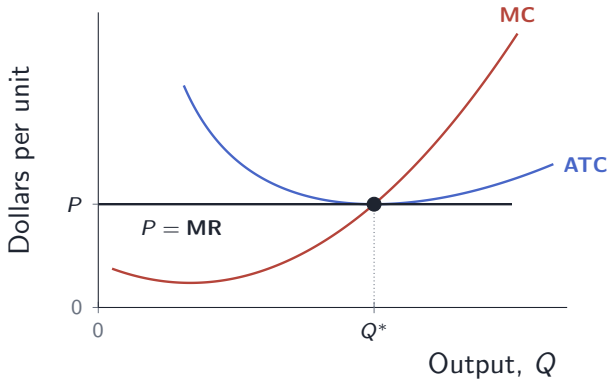
The firm in long-run equilibrium

$P = \text{MR}$: price taking.

$\text{MR} = \text{MC}$: the output rule.

$P = \text{ATC}$: zero economic profit.

MC meets ATC at its minimum: the efficient scale.

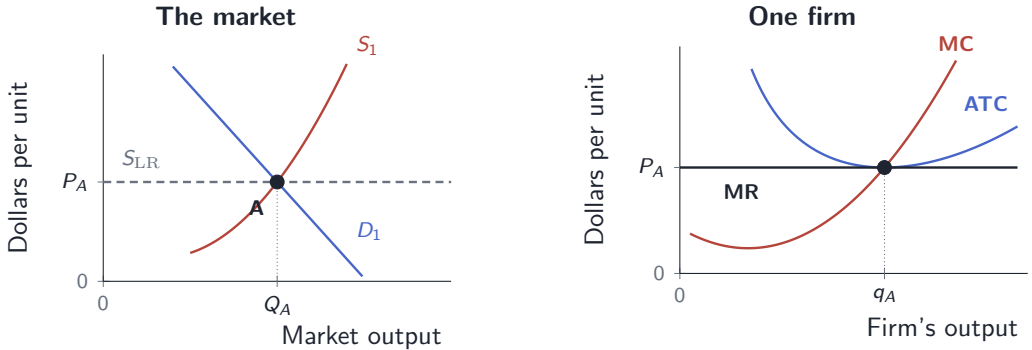


Finding

$P = \text{MR} = \text{MC} =$ minimum ATC. Revenue just covers total cost: $\text{TR} = P \times Q^* = \text{ATC} \times Q^* = \text{TC}$.

Market and firm at A

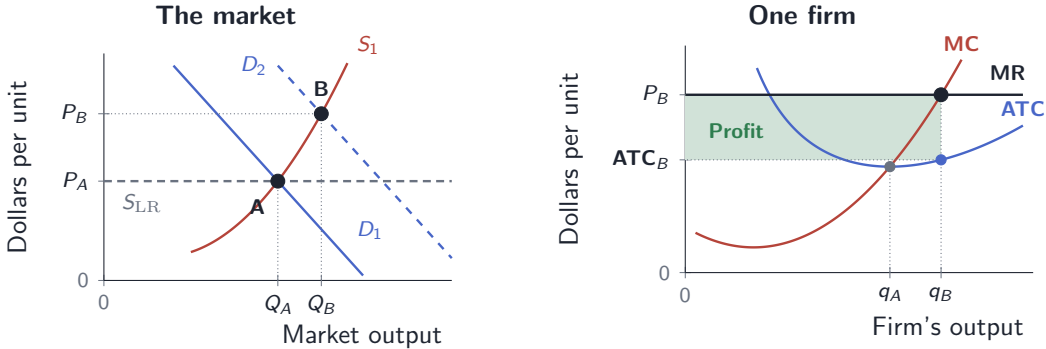
q: one firm; Q: market; N: number of firms. $Q = Nq$.



Finding

If price equals minimum ATC, each firm earns zero economic profit at efficient scale. At A this holds. S_{LR} is long-run market supply.

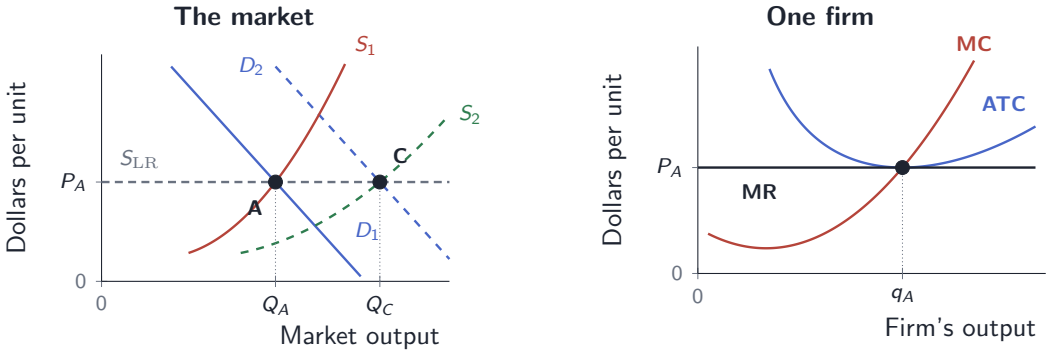
Market and firm at B



Finding

If demand rises with the number of firms fixed, price and each firm's output rise.
Profit rectangle: height $P_B - \text{ATC}_B$; width q_B; area $(P_B - \text{ATC}_B) \times q_B$.

Market and firm at C



Finding

If profit attracts entry, supply shifts from S_1 to S_2. Price returns to P_A.
Each firm returns to q_A and zero economic profit. Market output rises from Q_A to Q_C because there are more firms.

A fall in market demand

- In the short run, demand shifts left and the market price falls.
- Each operating firm reduces output along its MC curve.
- If price falls below minimum AVC, a firm shuts down. Identical firms share the same shutdown price.
- If losses persist, some firms exit. Market supply shifts left and price rises.

Finding

With identical firms and unchanged input prices, exit returns price to minimum ATC. There are fewer firms, each at its original efficient scale.

When long-run supply slopes up

- If expansion bids up the price of a scarce input, firms' costs rise. More output requires a higher market price.
- If firms have different costs, the lowest-cost firms enter first. More output requires higher-cost firms, which enter only at a higher price.

Finding

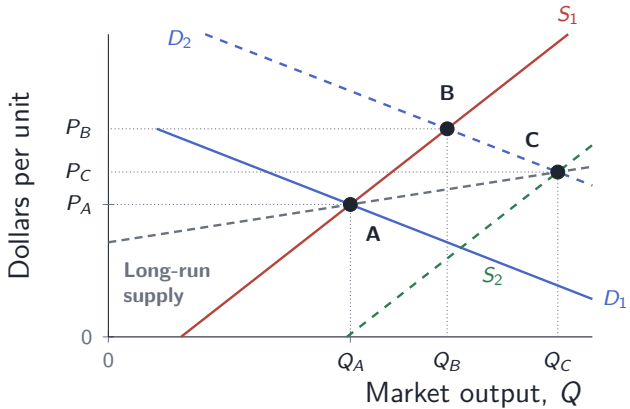
Free entry does not require horizontal long-run market supply. Long-run supply can slope upward.

An upward-sloping long-run supply curve

A: the initial long-run equilibrium.

B: demand rises. With the number of firms fixed, price rises along S_1.

C: entry shifts supply to S_2. Costs have risen, so price remains above P_A.



Finding

Entry lowers price from P_B to P_C. Higher costs keep P_C above P_A.

The marginal firm

Definition: Marginal firm

The firm that would exit in the long run if the price fell any lower. At the current price it earns zero economic profit.

- If costs differ, the long-run price equals the marginal firm's minimum ATC.
- Lower-cost firms can retain an economic profit if entrants cannot copy their cost advantage.
- Entry stops when no remaining entrant can earn a positive economic profit.

Why competition is efficient

- The market price measures what the last unit is worth to buyers.
- Each firm chooses $P = \text{MC}$: the last unit's value equals its cost.
- Producing more would cost more than buyers are willing to pay. Producing less would give up worthwhile trades.

Finding

With no external costs or benefits, the competitive market produces the efficient quantity.

Part 7 summary

- If demand rises, price and each firm's output first rise.
- If profit attracts entry, market supply shifts right and price falls.
- With unchanged costs, price returns to minimum ATC. Each firm returns to efficient scale; more firms make more market output.
- If input costs rise or new firms have higher costs, long-run supply can slope up.

Finding

Short run: firms change output. Long run: the number of firms also changes.

Your turn: Check your understanding

The graph shows a rise in demand, then adjustment from B to C.

1. What moves the market from B to C?

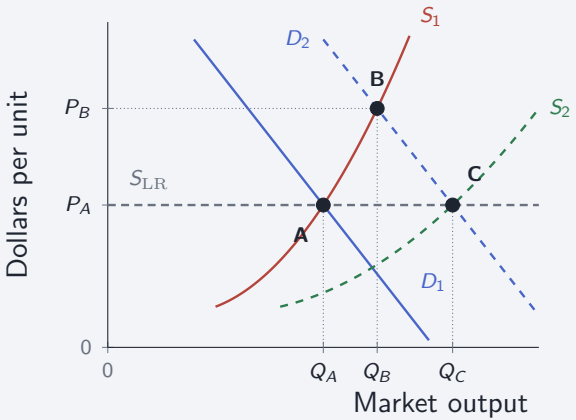
   A. Entry B. Exit C. Lower demand

2. Compared with P_A, price at C is:

   A. Higher B. The same C. Lower

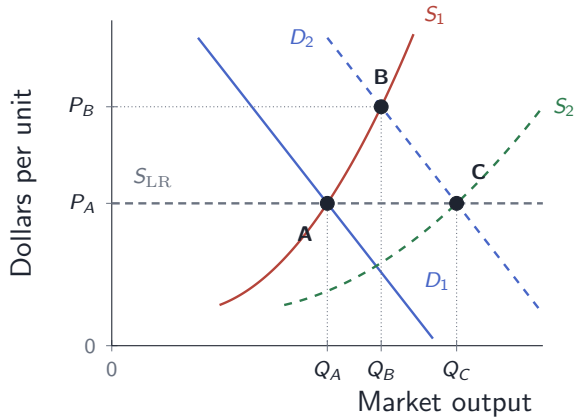
3. Rising input prices can make long-run supply slope up.

   A. True B. False



Solution: Check your understanding

1. **A.** Entry shifts supply right, from S_1 to S_2.
2. **B.** With unchanged costs, price returns to minimum ATC.
3. **A.** Higher input prices raise firms' costs.



Check: Part 7

When some resources used in production are only available in limited quantities, it is likely that the long-run supply curve in a competitive market is

A downward sloping.

B upward sloping.

C horizontal.

D vertical.

Check: Part 7 – answer

With some resources available only in limited quantities, long-run supply is likely

A downward sloping.

B upward sloping.

C horizontal.

D vertical.

Finding

Entry bids up the price of the scarce resource, so each firm's costs rise as the industry expands. More output then requires a higher price: long-run supply slopes up. It is horizontal only when costs do not change as firms enter.

Which rule answers which question

Question	Rule
What price does one firm face?	The market price: $P = \text{AR} = \text{MR}$.
How much should it produce?	$\text{MR} = \text{MC}$. Increase output if $\text{MR} > \text{MC}$; cut it if $\text{MC} > \text{MR}$.
Does it make a profit?	$\pi = (P - \text{ATC}) \times Q$.
Should it shut down now?	Shut down if $P <$ minimum AVC.
What is its short-run supply curve?	MC above minimum AVC: the rising part of MC.
Should it stay in the long run?	Exit if $P <$ minimum ATC.
What changes market supply?	Entry shifts it right; exit shifts it left.
Where do entry and exit stop?	$P =$ minimum ATC (identical firms, unchanged input prices).